

TABLE I: Resonance frequency and Q-factor of mode n in air and water, measured on a second device. The theoretic and experimental ratio's between the resonance frequencies upon immersion are $r_{\text{theory}} = f_{\text{water}}/f_{\text{vac}}$ and $r_{\text{exp}} = f_{\text{water}}/f_{\text{air}}$.

n	frequency (kHz)				Q - factor		
	air	water	r_{theory}	r_{exp}	air	water	water,theory
1	194.9	52.6	0.23	0.27	50	2.0	4.1
3	593.3	184	0.25	0.31	67	5.4	6.7

ratio between the frequencies:

$$\frac{f_{n,\text{water}}}{f_{n,\text{vac}}} = [1 + \frac{\pi w \rho}{4t \rho_c} \Gamma_R(\text{Re}, \kappa)]^{-0.5}, \quad (1)$$

where ρ_c denotes the average density of the gold coated resonator, Γ_R denotes the real part of $\Gamma(\text{Re}, \kappa)$, and $\pi w \rho / 4t \rho_c$ represents the mass loading parameter [10]. Although this model assumes a high Q-factor, experiments have shown it accurately describes the response of cantilever beams with Q-factors close to ours [11]. The results for the resonance frequencies and Q-factors are summarized in Table 1. We measured the resonance frequency from thermal noise spectra in atmospheric pressure and in the intrinsic damping regime in vacuum, and found $f_{1,\text{air}}/f_{1,\text{vac}} = 1.013$ and $f_{3,\text{air}}/f_{3,\text{vac}} = 1.007$, which is negligible compared to the shifts upon immersion in water [12]. We can thus assume $f_{n,\text{air}} = f_{n,\text{vac}}$, and the observed ratio's $f_{n,\text{water}}/f_{n,\text{vac}}$ are close to the theoretical prediction. The slightly higher prediction of the resonance frequency and Q-factors may be explained by the limited distance between the resonator and the substrate. In our experiment the distance is comparable to the resonator width, and in this case additional friction reduces the Q-factor by a factor of 2 when compared to a freely moving resonator [13].

To investigate the possibility to scale-down the dimensions of the string resonators, we have calculated the displacements and electromotive voltages at resonance [14]. In the harmonic regime, the tension force T is constant and sufficient to ensure string-like behavior with mode shapes and amplitudes N_n given by $u_n(x) = N_n \sin(\pi n x / L)$. The maximum displacement of the string for $n = 1, 3, \dots$, can then be calculated as [15]:

$$u_{\text{max},n} = \frac{4BIQ_n L^2}{n^3 \pi^3 T} \quad (2)$$

with B the magnetic field and I the current through the resonator. For currents in the mA-range and an estimated residual stress of 60 MPa, the amplitude at resonance is about $u_{\text{max},1} = 10$ nm for the first mode, and decreases with the mode number cubed. We note that the excellent thermalization in water allows large drive currents without damaging the resonator, and large resonator amplitudes can be obtained. The generated electromotive voltage is given by

$$V_{\text{EMF},n} = \frac{8}{\pi^3} \frac{f_n I B^2 Q_n L^3}{n^3 T} \quad (3)$$

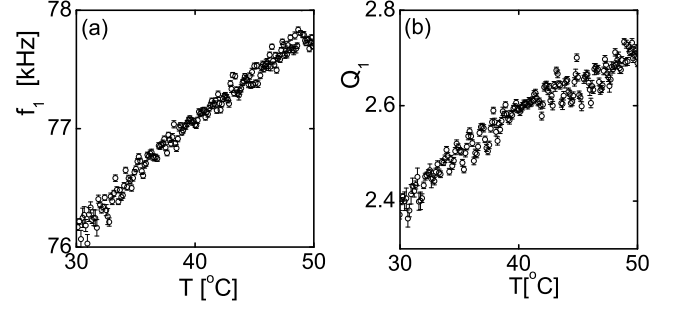


FIG. 3: Temperature dependence of the resonance frequency (a) and Q-factor (b) of the fundamental resonance mode of an immersed string resonator.

for odd mode numbers. For even n , the net flux rate is zero and no voltage is generated. The resonance frequency f_n scales with the inverse dimension (d^{-1}). Furthermore, in these experiments the tension force, T , results from residual stress and therefore scales as ($wt = d^2$). In a harmonic spectrum, $f_n = n f_1$, the generated voltage is thus independent on the mode number and proportional to the displacement $u_{n,\text{max}}$, which is adjustable by the driving force. Therefore $V_{\text{EMF}} \sim f L^3 / T \sim d^0$, independent on dimension.

Changes in liquid density and viscosity associated with temperature fluctuations pose a limit on the stability and accuracy of frequency-based measurements. Electrical and mechanical energy is dissipated by the strongly driven resonators (i.e. by ohmic resistance and by the viscous force), and an increase of water temperature is the result. The electrical and mechanical dissipation can be expressed as $U_{\text{in}} = I^2 R + 2\pi^2 m f_n^3 u_{\text{max}}^2 / Q$. In our experiments the contributions are on the order of $U_{\text{in},e} = 10^{-5}$ J/s for the electrical and $U_{\text{in},m} = 10^{-12}$ J/s for the mechanical dissipation, and electrical dissipation is dominant. If we apply the input energy to heat the $3 \mu\text{l}$ volume of water inside the flow cell, a temperature increase on the order of 60 mK/s is the result. We investigated the effect of such temperature fluctuations by measuring the resonance frequency and Q-factor during a controlled temperature sweep over a range of 20 °C. As shown in Fig. 3, both the resonance frequency and the Q-factor increase while increasing the temperature, as expected due to the reduction of the fluid density and viscosity. The results presented here for doubly clamped string resonators are comparable to results obtained earlier for cantilever beams [16]. The temperature dependence of the resonance frequency, approximately 100 Hz/°C in this experiment, underlines the need for accurate temperature control during mass sensing experiments in liquids.

In summary, we demonstrated magnetomotive drive and detection of micromechanical string resonators in water. The fundamental and third flexural resonance modes were detected, and the observed changes in resonance